

Experiment 5 – Lenses and Optical Instruments

Group 7

Biespel Pestañas, Giorgi Kipshidze

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Constructor University Bremen

Prof. Ph.D. Patrice Donfack

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Abstract

The aim of this experiment was to determine the focal lengths of different single lenses and lens systems, as well as to explore the basic operation of optical instruments such as a slide projector, a microscope, and telescopes. Using an optical bench, object and image distances were measured, and the focal lengths were calculated through the thin-lens equation and Bessel's method. Various optical instruments were then assembled from lenses with different focal lengths, and their magnifications were measured and compared with the corresponding theoretical values. The measured focal length of lens A was $f_A = [0.098]$ m and lens B was $f_B = [0.0763]$ cm. The Kepler telescope produced an inverted magnified image with an estimated magnification $M = [6.285]$, and the theoretical magnification of $M = [6.00]$. The slide projector produced an image with a magnification of $M = [8.956]$, which is close to the theoretical $M = [9.1]$. The microscope produced a magnified image of the object with magnification $M = [36.48]$ and the theoretical gave us $M = [35.6 \pm 0.1]$.

Introduction

Light is an electromagnetic wave, that when passing through lenses or mirrors, can be accurately described using geometrical optics as it also has properties of a particle. The property that light can be modeled as rays, traveling straight, and change direction when refracting or reflecting on surfaces such as mirrors or lenses. A lens uses refraction to bend these rays and form images. Depending on the focal length and position of an object, the lens can project a real (visible on a screen) or virtual image (seen by the eye but not captured on a screen).

These geometric optic models use trigonometric and geometric methods to describe lenses. A convex or converging lens bends parallel light rays, causing them to meet at a single point on the optical axis called the focal point. The distance between this point and the lens is the focal length. The relationship between the object distance g , image distance b , and **focal length f** is described by the thin lens equation:

$$\frac{1}{f} = \frac{1}{b} + \frac{1}{g}$$

eq. 1 – Thin Lens Equation

When two thin lenses are combined, their overall or **system focal length** can be found using

$$\frac{1}{f_{\text{sys}}} = \frac{1}{f_1} + \frac{1}{f_2}$$

eq. 2 – System of Focal Length Equation

Bessel's method offers a different way to find the focal length (Calculating focal lengths without knowing the exact lens center). By keeping the distance d between the object and the screen fixed and finding two positions of the lens that give sharp images, the focal length can be calculated as

$$f = \frac{d^2 - e^2}{4d}$$

eq. 3 – Bessel's Method Equation

The **magnification (M)** of a lens shows how much larger or smaller an image appears compared to the original object. It can be defined in terms of angular size as:

$$M = \frac{\text{angular size of virtual image } (\theta)}{\text{angular size of object } (\theta_0)}$$

eq. 4 – Magnification through Angular Size

The **magnification** of the projected image depends on the size of the image compared to the object.

$$M = \frac{B}{G} = \frac{b}{g} = \frac{b}{f} - 1$$

eq. 5 – Magnification through Distances and Focal Length

- **B** – size of the image on the screen
- **G** – size of the original slide
- **b** – image distance (distance from the lens to the screen)
- **g** – object distance (distance from the lens to the slide)
- **f** – focal length of the objective lens

A **microscope** uses two lenses — an objective and an eyepiece — to produce a magnified image of a very small object. The objective forms a real, enlarged image, which the eyepiece magnifies again to create a virtual image that can be comfortably viewed. The total magnification of a microscope is approximately:

$$M = M_{\text{ob}} \cdot M_{\text{oc}} \approx \frac{b_{\text{ob}}}{f_{\text{ob}}} \cdot \frac{25 \text{ cm}}{f_{\text{oc}}}$$

eq. 6 – Magnification of Microscope

Where b_{ob} is the image distance from the objective, and f_{ob} and f_{oc} are the focal lengths of the objective and eyepiece. When the image is projected onto a screen instead of being viewed directly, the magnification changes slightly:

$$M = M_{\text{ob}} \cdot M_{\text{oc}} \approx \frac{b_{\text{ob}}}{f_{\text{ob}}} \cdot \frac{b_{\text{oc}}}{f_{\text{oc}}}$$

eq. 7 - Magnification of Microscope Projected on Screen

A **Kepler telescope** works in a similar way, using two convex lenses. The **objective** creates a real, inverted image of a distant object, and the **eyepiece** magnifies that image. The angular magnification is given by

$$M_{\text{Kep}} = \frac{f_{\text{ob}}}{f_{\text{oc}}}$$

eq. 8 – Kepler Telescope Magnification

Experiment

Objective

The objective of this experiment is to investigate the basic properties of converging and diverging lenses and to study their use in optical instruments such as the microscope and telescope. The experiment aims to determine the focal lengths of different lenses using several methods and to analyze how lenses form real and virtual images depending on the object position. Additionally, the combination of lenses is examined to understand image magnification and focus on compound optical systems.

Equipment

- Optical bench with scale
- Screen and slide holder
- Illumination lamp or object (illuminated slide)

- Convex (converging) lenses of different focal lengths (e.g. +50 mm, +100 mm, +200 mm)
- Concave (diverging) lens (e.g. -100 mm)
- Lens holder and mountings
- Ruler or measuring tape
- Microscope tube and eyepiece
- Telescope tube and eyepiece

All measurements were performed on an optical bench, which allowed accurate alignment and positioning of the lenses, apertures, and screen along a common optical axis. This setup made it possible to study the properties of individual lenses and to build simple optical instruments such as a telescope, slide projector, and microscope.

A light source with an integrated condenser lens was used to produce a bright and almost parallel beam of light. Before starting the measurements, the room was darkened to make the images easier to see. The condenser was adjusted with the side slider until the filament of the lamp appeared sharply focused on a distant wall, ensuring even illumination across the object (the arrow slit) used during the experiment. All components — the lamp, object holder, lenses, and screen — were carefully aligned at right angles to the optical axis to reduce parallax and optical distortion.

For each task, the necessary optical elements were mounted on the bench and moved to the correct positions while their coordinates were recorded with millimeter precision. To determine the focal length, the distances between the object and image were measured for different lens positions. The uncertainty in each measurement was estimated from the scale resolution (± 1 mm) and from possible small alignment errors.

In the Bessel method, the distance between the object and screen was kept constant while the lens was moved between two positions where a sharp image appeared. The separation between these positions was then used to calculate the focal length. For combined lenses, both the lens equation and the Bessel method were applied to find the effective focal length of the system.

Uncertainties were determined using two approaches: error propagation for individual measurements, based on the uncertainties in the object distance (Δg) and image distance (Δb), and statistical analysis (mean value and standard deviation) for repeated measurements.

Error Analysis Methods

- 1) Mean Value

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- 2) Standard Deviation

$$\sigma_x = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- 3) error of the mean

$$\Delta \bar{x} = \frac{\sigma_x}{\sqrt{n}}$$

- 4) Error Propagation

$$\Delta y = \sqrt{\left(\frac{\partial y}{\partial x_1} \Delta x_1\right)^2 + \left(\frac{\partial y}{\partial x_2} \Delta x_2\right)^2 + \dots}$$

- 5) Relative Error

$$\delta = \frac{\Delta x}{x} \times 100\%$$

Presentation of Results

$$x = (\bar{x} \pm \Delta x)$$

Experiment 1: Determination of unknown focal lengths of convex lenses

The object (arrow slit) was placed close to the condenser, and the convex lens was used to project a sharp image onto the screen to determine its unknown focal length. During the

measurement, both the object distance (g) and image distance (b) were recorded for each lens position. The object remained fixed while the screen and lens were moved to obtain sharp images at different positions. The focal length (f) was then calculated using the thin lens equation

$$\frac{1}{f} = \frac{1}{g} + \frac{1}{b}$$

To estimate the uncertainty in f , several sets of measurements were performed under the same conditions, and the spread of results was evaluated using standard error analysis.

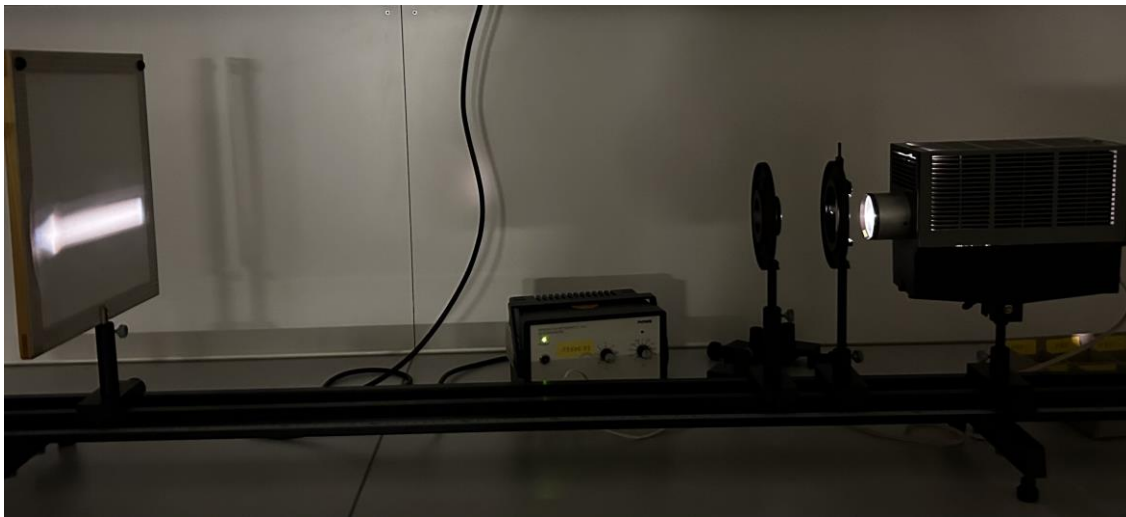


Fig 1-1 – Experiment 1 Set-up

Experiment 2: Determination of focal length of a lens system

For this part, we used the Bessel method to find the focal length of the lens system. The distance D between the object and the screen was kept fixed, and the lens was moved along the optical bench until a sharp image appeared on the screen. We noticed that there were two different lens positions that both gave clear images. These are the points where the object and image distances are swapped. We measured the distance between these two positions for each trial. Several readings were taken to calculate the mean and standard deviation of the lens positions, and we repeated the process for different values of D to check how consistent and precise our results were.

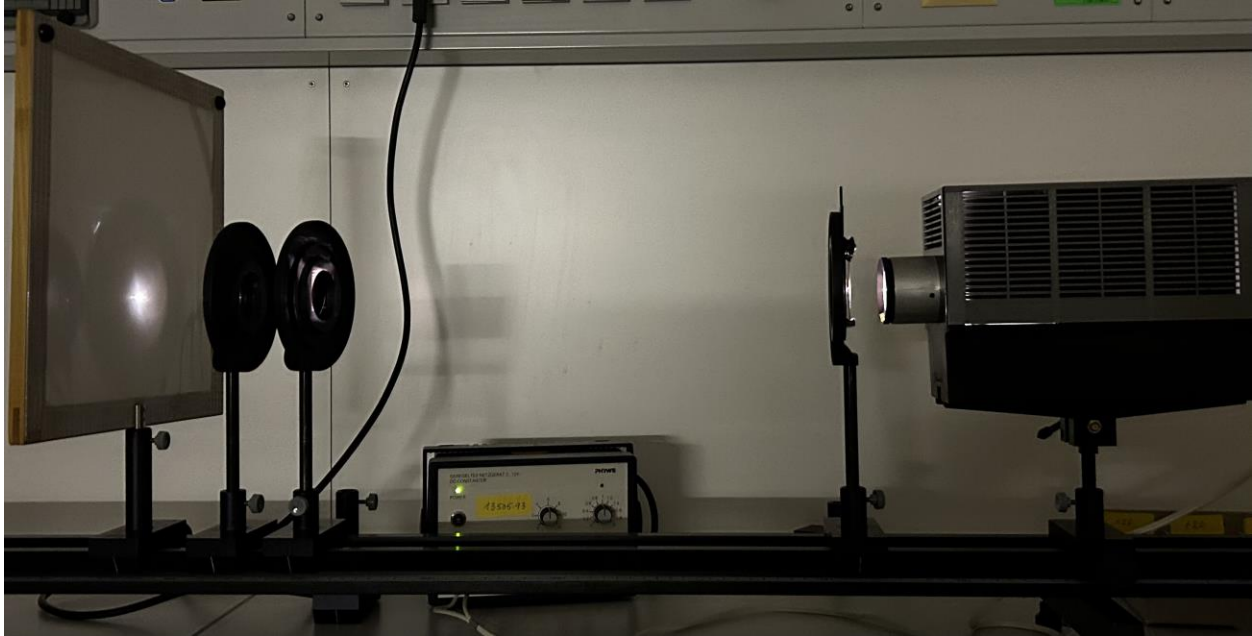


fig 2-1 - Experiment 2 Set-up

Experiment 3: Build up the telescope after Kepler

For the Kepler telescope, we used two convex lenses — one with a long focal length ($f_1 = 300\text{ mm}$) as the objective and another with a short focal length ($f_2 = 50\text{ mm}$) as the eyepiece. The lenses were mounted on the optical bench at a distance approximately equal to $f_1 + f_2$. When we looked through the short focal length lens, we could see an inverted and magnified image of distant objects, confirming the typical behavior of a Keplerian telescope.



fig 3-1 - Experiment 3 Viewpoint



fig 3-2 – Experiment 3 Set-up

Experiment 4: Set up a slide projector

In the slide projector experiment, the slide was placed close to the condenser, and an image was projected onto the screen using *Lens A and then using Lens B*. The condenser was adjusted so that the filament of the lamp was sharply focused onto the plane of the lens. This alignment provided even bright illumination of the slide, resulting in a clear and well-defined projected image.

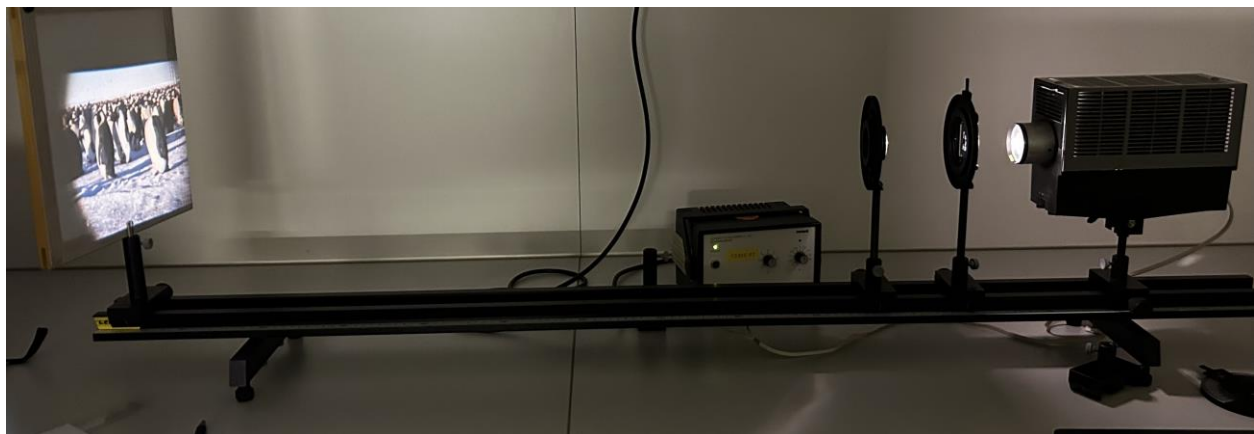


Fig 4-1 – Experiment 4 Set-up

Experiment 5: Set up a microscope

For the microscope setup, we used a short focal length objective lens L_1 ($f_1 = 20\text{ mm}$) to form an enlarged intermediate image of the object. A ground glass screen was placed at the position where this intermediate image came into focus so it could be clearly observed. The image was then viewed through an eyepiece L_2 ($f_2 = 20\text{ mm}$), which was adjusted so that the emerging rays were parallel to the eye, allowing comfortable viewing. To estimate the overall magnification, the size of the observed image was compared to a reference scale positioned at a standard viewing distance of 25 cm.



fig 5-1 – Experiment Set-up

Results

Task 1 – Determination of Unknown Focal Length of Convex Lenses

object Pa	I Lens PI	II Lens PII	image Pi	c	d	e	fsys	Mean fsys
1,13	1,03	0,909	0,79	0,04	0,34	0,121	234558823	0,076324150105736
1,13	1,02	0,96	0,827	0,04	0,303	0,06	2779702970	
1,13	1,01	0,965	0,833	0,04	0,297	0,045	2545454545	Standard Dev. σ
1,13	1	0,969	0,832	0,04	0,298	0,031	3693791946	0,00373859434428901
1,13	0,98	0,944	0,834	0,04	0,296	0,036	2905405405	
1,13	0,96	0,947	0,823	0,04	0,307	0,013	5612377850	Error Δ fsys
1,13	0,95	0,946	0,82	0,04	0,31	0,004	7487096774	0,0011822473375377
1,13	0,94	0,938	0,814	0,04	0,316	0,002	8996835443	
1,13	0,93	0,927	0,807	0,04	0,323	0,003	0743034055	Error Propagation
1,13	0,92	0,917	0,797	0,04	0,333	0,003	3243243243	0,00638770697547404
Focal Length				0,0763	$\pm$	0,0012	m	

For task 1, the goal is to determine the focal length for a convex lens. This is done through the Lens Maker equation and requires the distances between the object and the lens g , and the screen and the lens b . Taking the positions of each component – Screen, Lens, and object (arrow) – the difference of these positions will be the distance.

Lens (PI)	Arrow (Pa)	Image (Pi)	g	b	f A	$1/g$	$1/b$	$1/f$
1.01	1.13	0.5	0.12	0.51	0.097143	8.333333	1.960784	10.29412
1	1.13	0.51	0.13	0.49	0.102742	7.692308	2.040816	9.733124
0.99	1.13	0.57	0.14	0.42	0.105	7.142857	2.380952	9.52381
0.98	1.13	0.65	0.15	0.33	0.103125	6.666667	3.030303	9.69697
0.97	1.13	0.695	0.16	0.275	0.101149	6.25	3.636364	9.886364
0.96	1.13	0.72	0.17	0.24	0.099512	5.882353	4.166667	10.04902
0.95	1.13	0.73	0.18	0.22	0.099	5.555556	4.545455	10.10101
0.94	1.13	0.73	0.19	0.21	0.09975	5.263158	4.761905	10.02506
0.93	1.13	0.731	0.2	0.199	0.099749	5	5.025126	10.02513
0.92	1.13	0.732	0.21	0.188	0.099196	4.761905	5.319149	10.08105
0.46	1.13	0.348	0.67	0.112	0.095959	1.492537	8.928571	10.42111
0.42	1.13	0.31	0.71	0.11	0.095244	1.408451	9.090909	10.49936
0.4	1.13	0.29	0.73	0.11	0.095595	1.369863	9.090909	10.46077
0.38	1.13	0.271	0.75	0.109	0.095169	1.333333	9.174312	10.50765
0.37	1.13	0.261	0.76	0.109	0.095328	1.315789	9.174312	10.4901

Table 1-1 – Position, Distance, Focal Length Data

The data was treated by calculating the mean, standard deviation, and statistical error.

Mean	0.098910783
Standard Deviation	0.003156648
Error of the Mean	0.012225644

Table 1-2 – Mean and Error of Convex Lens’ Focal Length

After calculating the Focal Length, the propagated error is calculated from the partial derivatives of function f with values from the data.

$$\frac{\delta f}{\delta g} * \Delta g = \frac{b^2}{(b+g)^2} * \Delta g$$

Partial Derivative off with respect to g

$$\frac{\delta f}{\delta b} * \Delta b = \frac{g^2}{(b+g)^2} * \Delta b$$

Partial Derivative with respect to b

$$\Delta f_{avg} = \sqrt{\left(\frac{\delta f}{\delta g} \cdot \Delta g\right)^2 + \left(\frac{\delta f}{\delta b} \cdot \Delta b\right)^2}$$

Propagated Error of f

Δg	0.0001
Δb	0.0001
Δf_{avg}	3.81494E-06

Task 2 determining the focal length

f Bessel = 0,0769230769230769

object Pa	I Lens PI	II Lens PII	image Pi	c	d	e	fsys	Mean fsys	
1,13	1,03	0,909	0,79	0,04	0,34	0,121	0,074234558823529 4	0,076324150105736	
1,13	1,02	0,96	0,827	0,04	0,303	0,06	0,072779702970297		
1,13	1,01	0,965	0,833	0,04	0,297	0,045	0,072545454545454 5	Standard Dev. σ	
1,13	1	0,969	0,832	0,04	0,298	0,031	0,073693791946308 7	0,003738594344289 01	

1,13	0,98	0,944	0,834	0,04	0,296	0,036	0,072905405405405 4		
1,13	0,96	0,947	0,823	0,04	0,307	0,013	0,076612377850162 9	Error Δfsys	
1,13	0,95	0,946	0,82	0,04	0,31	0,004	0,077487096774193 6	0,001182247337537 7	
1,13	0,94	0,938	0,814	0,04	0,316	0,002	0,078996835443038		
1,13	0,93	0,927	0,807	0,04	0,323	0,003	0,080743034055727 6	Error Propagation	
1,13	0,92	0,917	0,797	0,04	0,333	0,003	0,083243243243243 3	0,006387706975474 04	
	Focal Length		0,0763	±	0,0012	m			

Statistical Error and Error Propagation Comparison

Statistical Error is smaller as it compares various samples while the Propagated Error is bigger because it only has one.

Task 3 - Kepler Telescope

The best combination, 300 and 50, gave a clear and detailed image of the object from far away. Compared to theoretical magnification, it is exactly equal as the lenses are made with the exact specifications. The only error would be if the lens had imperfections.

$f_{\text{obj}} = 300 \text{ mm}$

$f_{\text{eye}} = 50 \text{ mm}$

$M(\text{theoretical}) = |-300/50| = 6 \text{ mm}$

Experimental

Image (m)= 2,2

Object (m)= 0,35

						Experimental		
	Theoretical Magnification					M		Error
	6					6,28571428571429	±	0,1
								m

The measured magnification of the Kepler telescope was consistent with the theoretical value within the experimental uncertainty. As expected, the image appeared inverted, and some loss of brightness occurred due to light absorption and reflection in the optical system

Task 4 - Slide projector

Sketch.



The single most important optical part of the projector: The Lens

- Test types of these parts:
- Type 1 (A)
- Type 2 (B)

Comparison: The Type A lens produced a clearer image closer up. Type B produced a clearer image farther.

Describe and characterize the image size and quality on the final setup (e.g. brightness, sharpness, distortions,...).

The image size and quality were clear on the screen

Image Size	Object Size
0,206	0,023

Experimental magnification = $8.95652173913043 \pm 0.0001$ m

Theoretical Magnification = 9.1

Comparison: The Magnification was similar but out of the error range.

Task 5 - Set Up a Microscope

Sketch.



b_obj	f_obj	b_oc	f_oc	M
0,046	0,02	0,793	0,05	36,478

M = 36.48 (Theoretical)

M = 35.60 ± 0.10 (Experimental)

$$\frac{b_{obj}}{f_{obj}} * \frac{b_{oc}}{f_{oc}} = M_{ob} * M_{oc} = M$$

image	object	Experimental		
0,178	0,005	35,6	±	0,1

Without moving the screen or the lens, the image became blurrier when the lens was switched. When switched and moved, the image became smaller or bigger based on the magnification.

Discussion and Conclusion

In this experiment, we explored how lenses form images and how they can be combined to build simple optical instruments. Our measurements of focal lengths and magnifications were mostly consistent with the theoretical values, which shows that the methods we used were reliable. The small differences we observed can likely be explained by practical limitations, such as slight misalignments in the setup or the difficulty of deciding exactly when an image appears perfectly sharp. Overall, the results support the underlying optical principles and gave us a clearer understanding of how devices like microscopes and telescopes work in practice.